

PicoCTF Write-Up

Cooking solutions, one flag at a time.

SSH & Netcat | General Skills | Beginner | June 2026

■ Introduction

I'm CYB3R_CH3F — I cook solutions. As part of my cybersecurity journey, I'm solving PicoCTF challenges and documenting every step so that even someone with zero knowledge can follow along. This write-up covers two essential tools: **SSH** and **Netcat**. Master these and you've got two key ingredients in your kitchen.

GENERAL SKILLS

EASY

NETWORKING

BEGINNER

■ Challenge 1 — Secure Shell (SSH)

What's the challenge asking?

Connect to a remote server using SSH at a custom port, with a given username and password. The flag is printed directly in the session output once you log in.

■ PicoCTF Challenge Page — Secure Shell (SSH)

General Skills Easy 95,662 solves

by Jeffery John

shell ssh browser_webshell_solvable

Using a Secure Shell (SSH) is going to be pretty important.
Can you ssh as ctf-player to titan.picoctf.net at port 60126 to get the flag?
You'll also need the password 84b12bae. If asked, accept the fingerprint with yes.
If your device doesn't have a shell, you can use: <https://webshell.picoctf.org>
If you're not sure what a shell is, check out our Primer: https://primer.picoctf.com/#_the_shell

INSTANCE

Expires in 11:56

Restart Instance

Hints ⓘ

Reveal hint 1

Reveal hint 2

Reveal hint 3

Reveal hint 4

Concept: What is SSH?

SSH (Secure Shell) is an encrypted protocol for remote login. Think of it as a locked tunnel into another computer — everything you send is encrypted. The default port is 22, but servers can run SSH on any port.

Connection Details

Host	titan.picoctf.net
Port	60126
Username	ctf-player
Password	84b12bae

Command Used

```
$ ssh -p 60126 ctf-player@titan.picoctf.net
```

-p 60126	-> custom port (default SSH port is 22)
ctf-player@...	-> username @ hostname
password	-> 84b12bae

My Thought Process

- Spotted the custom port **60126** — means I need the **-p** flag.
- Got a fingerprint warning on first connect — typed **yes** to accept it.
- Entered the password. Flag came back immediately in the welcome message.

Terminal — SSH login, fingerprint accepted, flag captured

```
(kali㉿kali)-[~]
$ ssh -p 60126 ctf-player@titan.picoctf.net

The authenticity of host '[titan.picoctf.net]:60126 ([3.139.174.234]:60126)' can't be established.
ED25519 key fingerprint is: SHA256:4S9EbTSSRZm32I+cdM5TyzthpQryv5kudRP9PIKT7XQ
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[titan.picoctf.net]:60126' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
ctf-player@titan.picoctf.net's password:
Welcome ctf-player, here's your flag: picoCTF{s3cur3_c0nn3ct10n_07a987ac}
Connection to titan.picoctf.net closed.

(kali㉿kali)-[~]
$
```

Flag Captured

```
picoCTF{s3cur3_c0nn3ct10n_07a987ac}
```

Key Takeaways

- > SSH = **encrypted** remote login. Nothing travels in plaintext.
- > Always use **-p** when the port is not the default 22.
- > Fingerprint prompt = SSH protecting you from man-in-the-middle attacks.

Challenge 2 — Netcat (nc)

What's the challenge asking?

Connect to a server using Netcat at a given host and port. The server sends the flag back — just read it.

what's a net cat?

General Skills **Easy** 146,901 solves
by Sanjay C/Danny Tunitis

Using netcat (nc) is going to be pretty important. Can you connect to `fickle-tempest.picoctf.net` at port 59546 to get the flag?

INSTANCE

Expires in 29:56

[Restart Instance](#)

Concept: What is Netcat?

Netcat (nc) is called the **Swiss Army knife of networking**. It opens a raw TCP or UDP connection with no encryption and no login — just a direct line. Whatever the server sends, you see it. In CTFs, nc is used constantly.

Connection Details

Host	<code>fickle-tempest.picoctf.net</code>
Port	<code>59546</code>

Mistake I Made (So You Don't)

I copied the challenge description text directly and pasted it as a command. The description said 'at port 59546' in plain English, so those words ended up in my terminal:

```
X $ nc fickle-tempest.picoctf.net at port 59546
invalid port at
```

The words 'at port' are NOT part of the command.
Netcat just takes the host then the port number directly.

The correct command:

```
OK $ nc fickle-tempest.picoctf.net 59546
```

```
nc          -> netcat utility
hostname    -> the server
59546       -> port number only – nothing else
```

■ Terminal — wrong command error then correct command + flag

```
(kali㉿kali)-[~]
$ nc fickle-tempest.picoctf.net at port 59546
invalid port at

(kali㉿kali)-[~]
$ nc fickle-tempest.picoctf.net 59546
You're on your way to becoming the net cat master
picoCTF{nEtCat_Mast3ry_5c7cC1a9}

(kali㉿kali)-[~]
$
```

My Thought Process

- First run failed — pasted the description text word for word. Classic mistake.
- Realised nc syntax is just **nc <host> <port>**. Stripped the extra words.
- Second run worked instantly. Server printed the flag.

■ Flag Captured

picoCTF{nEtCat_Mast3ry_5c7cC1a9}

Key Takeaways

- > Netcat syntax: **nc <host> <port>** — port number only, no extra words.
- > No encryption, no login. Whatever the server sends, you get it raw.
- > **Read the actual command syntax, not the plain English description.**

■ SSH vs Netcat — Quick Comparison

	SSH	Netcat (nc)
Encryption	Yes	No
Auth	Username + Password / Key	None
Syntax	ssh -p PORT user@host	nc host PORT
CTF use	Log into remote machines	Talk to challenge servers
Traffic	Encrypted tunnel	Raw plaintext

GitHub

github.com/cyb3r-ch3f

LinkedIn

linkedin.com/in/adebakin-ifeoluwa

X / Twitter

x.com/jst_ify